Machine Learning

Handwritten-Style Student Notes
From Basics to Practical Model Building

Learning goal: Understand how machines learn patterns from data and how to build, evaluate, and improve ML models.

Important idea: Machine Learning is a branch of Artificial Intelligence in which a computer learns patterns from data instead of being explicitly programmed for every rule.

1. What is Machine Learning?

Definition

Machine Learning is a technique that allows computers to learn from historical data and make predictions or decisions on new data.

Real-world examples

Email spam detection, house-price prediction, movie recommendations, fraud detection, face recognition, and customer churn prediction.

Traditional programming vs ML

Traditional: Rules + Data → Output. ML: Data + Output examples → Model/Rules.

Basic workflow

Collect data → Clean data → Explore data → Prepare features → Split data → Train model → Evaluate model → Deploy model.

2. Types of Machine Learning

Supervised Learning

The model learns from labeled data. Each input has a known output. Examples: house-price prediction and disease classification.

Unsupervised Learning

The model receives data without target labels and discovers hidden patterns. Examples: customer segmentation and grouping similar products.

Reinforcement Learning

An agent learns by interacting with an environment and receiving rewards or penalties. Examples: robotics and game-playing agents.

3. Supervised Learning

Regression

Predicts a continuous numerical value. Examples: salary, temperature, sales, and house price.

Classification

Predicts a category or class. Examples: spam/not spam, pass/fail, fraud/not fraud.

Features and target

Features (X) are input columns used for prediction. Target (y) is the output column that the model learns to predict.

4. Data Preprocessing

Why preprocessing?

Raw data may contain missing values, duplicate rows, inconsistent formats, outliers, and categorical text.

Common steps

1. Remove duplicates. 2. Handle missing values. 3. Convert data types. 4. Encode categorical columns. 5. Scale numerical features. 6. Detect outliers.

Missing values

Numerical values can be filled using mean or median. Categorical values can be filled using mode or a separate category such as 'Unknown'.

Encoding

Label encoding maps categories to numbers. One-hot encoding creates separate 0/1 columns for each category.

Data leakage

Data leakage occurs when information from the target or future data accidentally enters the training features. It can produce unrealistically high performance.

5. Train-Test Split

Meaning

Training data is used to learn patterns. Testing data is used to check how well the model performs on unseen data.

Typical split

80% training and 20% testing is common, but the correct split depends on the dataset and problem.

Python syntax

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

random_state

It makes the split reproducible, so the same code gives the same split each time.

6. Feature Scaling

Why scaling?

Some algorithms are affected by feature magnitude. For example, income may be measured in thousands while age is measured in years.

Standardization

Transforms values so that the feature has approximately mean 0 and standard deviation 1.

Formula

z = (x − mean) / standard deviation

Normalization

Often means Min-Max scaling, which converts values into a fixed range, commonly 0 to 1.

StandardScaler code

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Important rule

Fit the scaler only on training data. Use transform on test data to avoid data leakage.

7. Linear Regression

Definition

Linear Regression predicts a continuous target using a linear relationship between input features and output.

Equation

$y = mX + c$

Multiple linear regression

$y = b0 + b1X1 + b2X2 + ... + bnXn$

Use cases

House-price prediction, sales forecasting, salary prediction, and demand estimation.

Python code

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

8. Logistic Regression

Definition

Logistic Regression is a classification algorithm that predicts the probability of a class.

Output

The sigmoid function converts a score into a value between 0 and 1.

Use cases

Pass/fail, customer churn, fraud detection, and disease-risk classification.

Important point

Although its name contains regression, it is commonly used for classification.

9. K-Nearest Neighbors (KNN)

Definition

KNN predicts the class of a new point by looking at the nearest K data points.

Steps

1. Select K. 2. Calculate distances. 3. Find nearest neighbors. 4. Use majority vote for classification or average for regression.

Important point

KNN is distance-based, so feature scaling is usually important.

Choosing K

Small K may overfit and large K may underfit. Try several values and compare validation performance.

10. Decision Tree

Definition

A Decision Tree makes predictions by asking a sequence of questions about features.

Parts

Root node, decision/internal nodes, branches, and leaf nodes.

Advantages

Easy to understand, handles numerical and categorical data with suitable preprocessing, and can model non-linear relationships.

Overfitting

A very deep tree may memorize the training data. Control it using max_depth, min_samples_split, or pruning.

11. Random Forest

Definition

Random Forest combines many decision trees and combines their predictions.

For classification

Uses majority voting.

For regression

Uses the average of tree predictions.

Advantages

Usually more stable than one decision tree and can model complex relationships.

Important hyperparameters

n_estimators, max_depth, min_samples_split, max_features.

12. Support Vector Machine (SVM)

Definition

SVM finds a decision boundary that separates classes with the largest possible margin.

Kernel

A kernel can help SVM model non-linear boundaries. Common kernels include linear, polynomial, and RBF.

Important point

Scaling is generally important for SVM, especially with RBF and polynomial kernels.

13. Naive Bayes

Definition

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem.

Assumption

It assumes features are conditionally independent given the class.

Use cases

Text classification, spam detection, sentiment analysis, and document categorization.

14. Clustering — K-Means

Definition

K-Means groups similar data points into K clusters.

Steps

1. Select K. 2. Initialize centroids. 3. Assign points to nearest centroid. 4. Recalculate centroids. 5. Repeat until stable.

Use cases

Customer segmentation, image compression, and grouping similar products.

Choosing K

The elbow method compares within-cluster variation for different K values.

15. Model Evaluation Metrics

Regression metrics

MAE: average absolute error. MSE: average squared error. RMSE: square root of MSE. R^2: proportion of target variance explained by the model.

Classification metrics

Accuracy = correct predictions / total predictions.

Precision

Of all predicted positive cases, how many were actually positive?

Recall

Of all actual positive cases, how many did the model correctly identify?

F1-score

Harmonic mean of precision and recall.

Confusion matrix

Contains True Positive, True Negative, False Positive, and False Negative counts.

16. Overfitting and Underfitting

Overfitting

The model performs very well on training data but poorly on unseen data. It has learned noise or memorized examples.

Underfitting

The model is too simple and performs poorly on both training and testing data.

Solutions

Use cross-validation, regularization, more data, feature selection, early stopping, or simpler models.

17. Cross-Validation

Definition

Cross-validation divides the training data into multiple parts and trains/validates the model several times.

K-Fold method

In K-Fold cross-validation, the data is divided into K folds. Each fold becomes the validation set once.

Benefit

It provides a more reliable estimate of model performance than one single split.

18. End-to-End ML Pipeline

Project steps

Understand business problem → Collect data → Clean data → EDA → Feature engineering → Split data → Build baseline → Train models → Tune hyperparameters → Evaluate → Save model → Deploy → Monitor.

Business example

For a loan prediction system, the goal may be to predict whether an applicant is likely to repay a loan. Features can include income, credit history, loan amount, and employment information.

19. Common Python Libraries

NumPy

Numerical arrays and mathematical operations.

Pandas

Data loading, cleaning, filtering, grouping, and transformation.

Matplotlib

Data visualization.

Seaborn

Statistical visualization.

Scikit-learn

Preprocessing, ML algorithms, metrics, pipelines, and model selection.

Joblib/Pickle

Saving and loading trained Python models; use carefully and only with trusted files.

20. Quick Interview Revision

What is ML?

A method that learns patterns from data to make predictions or decisions.

What is the difference between classification and regression?

Classification predicts categories; regression predicts continuous numerical values.

Why split data?

To evaluate generalization on unseen data.

Why scale features?

To prevent large-magnitude features from dominating distance- or gradient-based algorithms.

What is overfitting?

High training performance but poor generalization.

What is a hyperparameter?

A setting chosen before or during training, such as K in KNN or max_depth in a tree.

What is data leakage?

Using information during training that would not be available when making real predictions.

What is a baseline model?

A simple reference model used to compare whether more complex models provide improvement.